



April 11, 2025

Oncospace, Inc.
Sigrid Schoepel
Regulatory Affairs
1812 Ashland Ave., Suite 100K
Baltimore, Maryland 21205

Re: K242748

Trade/Device Name: Oncospace
Regulation Number: 21 CFR 892.5050
Regulation Name: Medical Charged-Particle Radiation Therapy System
Regulatory Class: Class II
Product Code: MUJ
Dated: March 12, 2025
Received: March 12, 2025

Dear Sigrid Schoepel:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device"

(<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory->

[assistance/contact-us-division-industry-and-consumer-education-dice](#)) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

A handwritten signature in black ink, appearing to read "Lora D. Weidner". The signature is written in a cursive style. A large, light blue "FDA" watermark is visible in the background behind the signature.

Lora D. Weidner, Ph.D.
Assistant Director
Radiation Therapy Team
DHT8C: Division of Radiological
Imaging and Radiation Therapy Devices
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

Submission Number (if known)

K242748

Device Name

Oncospace

Indications for Use (Describe)

Oncospace is used to configure and review radiotherapy treatment plans for a patient with malignant or benign disease in the head and neck, thoracic, abdominal, and pelvic regions. It allows for set up of radiotherapy treatment protocols, association of a potential treatment plan with the protocol(s), submission of a dose prescription and achievable dosimetric goals to a treatment planning system, and review of the treatment plan. It is intended for use by qualified, trained radiation therapy professionals (such as medical physicists, oncologists, and dosimetrists). This device is for prescription use by order of a physician.

Type of Use (Select one or both, as applicable)



Prescription Use (Part 21 CFR 801 Subpart D)



Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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510(k) Summary

I. SUBMITTER

Oncospace, Inc.
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Email: Sigrid.Schoepel@oncospac.com
Contact Person: Sigrid Schoepel

Date Prepared: 2025 March 7

II. DEVICE

Name of Device: Oncospace

Common or Usual Name: System, Planning, Radiation Therapy Treatment

Classification Name: Medical charged-particle radiation therapy system (21 CFR 892.5050)

Regulatory Class: II

Product Code: MUJ

III. PREDICATE DEVICE

Oncospace, K222803

This predicate device has not been the subject of a recall.

IV. DEVICE DESCRIPTION

The Oncospace software supports radiation oncologists and medical dosimetrists during radiotherapy treatment planning. The software includes locked machine learning algorithms. During treatment planning, the Oncospace software works in conjunction with, and does not replace, a treatment planning system (TPS).

The Oncospace software is intended to augment the treatment planning process by:

- allowing the radiation oncologist to select and customize a treatment planning protocol that includes dose prescription (number of fractions, dose per fraction, dose normalization), a delivery method (beam type and geometry), and protocol-based dosimetric goals/objectives for treatment targets, and organs at risk (OAR);
- predicting dosimetric goals/objectives for OARs based on patient-specific anatomical geometry;

- automating the initiation of plan optimization on a TPS by supplying the dose prescription, delivery method, protocol-based target objectives, and predicted OAR objectives;
- providing a user interface for plan evaluation against protocol-based and predicted goals.

Diagnosis and treatment decisions occur prior to treatment planning and do not involve Oncospace. Decisions involving Oncospace are restricted to setting of dosimetric goals for use during plan optimization and plan evaluation. Human judgement continues to be applied in accepting these goals and updating them as necessary during the iterative beam optimization process. Human judgement is also still applied as in standard practice during plan quality assessment; the protocol-based OAR goals are used as the primary means of plan assessment, with the role of the predicted goals being to provide additional information as to whether dose to an OAR may be able to be further lowered.

When Oncospace is used in conjunction with a TPS, the user retains full control of the TPS, including finalization of the treatment plan created for the patient. Oncospace also does not interface with the treatment machines. The risk to patient safety is lower than a TPS since it only informs the treatment plan, does not allow region of interest editing, does not make treatment decisions, and does not interface directly with the treatment machine or any record and verify system.

Oncospace's OAR dose prediction approach, and the use of predictions in end-to-end treatment planning workflow, has been tested for use with a variety of cancer treatment plans. These included a wide range of target and OAR geometries, prescriptions and boost strategies (sequential and simultaneous delivery). Validity has thus been demonstrated for the range of prediction model input features encountered in the test cases. This range is representative of the diversity of the same feature types (describing target-OAR proximity, target and OAR shapes, sizes, etc.) encountered across all cancer sites. Given that the same feature types will be used in OAR dose prediction models trained for all sites, the modeling approach validated here is not cancer site specific, but rather is designed to predict OAR DVHs based on impactful features common to all sites. The software is designed to be used in the context of all forms of intensity-modulated photon beam radiotherapy. The planning objectives themselves are intended to be TPS-independent: these are instead dependent on the degree of organ sparing possible given the beam modality and range of delivery techniques for plans in the database. To facilitate streamlined transmission of DICOM files and plan parameters Oncospace includes scripts using the treatment planning system's scripting language (for example, Pinnacle).

The Oncospace software includes an algorithm for transforming non-standardized OAR names used by treatment planners to standardized names defined by AAPM Task Group 263. This matching process primarily uses a table of synonyms that is updated as matches are made during use of the product, as well as a Natural Language Processing (NLP) model that attempts to match plan names not already in the synonym table. The NLP model selects the most likely match, which may be a correct match to a standard OAR name, an incorrect match, or no match (when the model considers this to be most likely, such as for names resembling a target). The user can also manually match names using a drop-down menu of all TG-263 OAR names. The user is instructed to check each automated match and make corrections using the drop-down menu as needed.

V. INDICATIONS FOR USE

Oncospace is used to configure and review radiotherapy treatment plans for a patient with malignant or benign disease in the head and neck, thoracic, abdominal, and pelvic regions. It allows for set up of radiotherapy treatment protocols, association of a potential treatment plan with the protocol(s), submission of a dose prescription and achievable dosimetric goals to a treatment planning system, and review of the treatment plan. It is intended for use by qualified, trained radiation therapy professionals (such as medical physicists, oncologists, and dosimetrists). This device is for prescription use by order of a physician.

VI. COMPARISON OF TECHNOLOGICAL CHARACTERISTICS WITH THE PREDICATE DEVICE

The Oncospace subject device is a software-only medical device that performs the same functions as the Oncospace predicate device. The following differences exist between the subject and predicate devices:

- The prostate, head, and neck models have been updated to improve performance.
- The thoracic, abdominal, and gynecological regions have been added using the same machine learning methods as in the predicate device.

Note: The subject device algorithms, and any future algorithm updates, are locked prior to clinical use.

Element	Subject	Predicate	Conclusion
Device Name	Oncospace	Oncospace	Identical
510(k) Owner	Oncospace, Inc.	Oncospace, Inc.	
510(k) Number	--	K222803	
Product Code	MUJ	MUJ	Identical
Product Name	System, Planning, Radiation Therapy Treatment	System, Planning, Radiation Therapy Treatment	Identical
Intended Use	Oncospace is used to configure and review radiotherapy treatment plans for a patient with malignant or benign disease in the head and neck, thoracic, abdominal, and pelvic regions. It allows for set up of radiotherapy treatment protocols, association of a potential treatment plan with the protocol(s), submission of a dose prescription and achievable dosimetric goals to a treatment planning system, and review of the treatment plan. It is intended for use by qualified, trained radiation therapy professionals (such as medical physicists, oncologists, and dosimetrists).	Oncospace is used to configure and review radiotherapy treatment plans for a patient with malignant or benign disease in the prostate, head, and neck regions. It allows for set up of radiotherapy treatment protocols, association of a potential treatment plan with the protocol(s), submission of a dose prescription and achievable dosimetric goals to a treatment planning system, and review of the treatment plan. It is intended for use by qualified, trained radiation therapy professionals (such as medical physicists, oncologists, and dosimetrists). This device is for prescription use by order of a physician.	Substantially Equivalent Added thoracic and abdominal regions, added gynecological sites to expand the prostate reference to the pelvic region.

Element	Subject	Predicate	Conclusion
	This device is for prescription use by order of a physician.		
Operating System	Windows/Web-browser	Windows/Web-browser	Identical
Platform	Client-Server (Clinic-provided client machines, cloud Windows servers controlled by Oncospace)	Client-Server (Clinic-provided client machines, cloud Windows servers controlled by Oncospace)	Identical
DICOM-RT Compliant	Yes	Yes	Identical
Full Treatment Planning System	No	No	Identical
Connected to or Controlling of Radiation Delivery Devices	No	No	Identical
Typical Users	Medical professionals, including but not limited to, radiation oncologists, medical physicists or physicians.	Medical professionals, including but not limited to, radiation oncologists, medical physicists or physicians.	Identical
Patient Population	There are no demographic, regional, or cultural limitations for patients. It is up to the user to determine if the system can be used for a patient.	There are no demographic, regional, or cultural limitations for patients. It is up to the user to determine if the system can be used for a patient.	Identical
Body Regions	Head and neck Thoracic Abdominal Pelvic The thoracic, abdominal, and gynecological regions have been added using the same machine learning methods as in the predicate device.	Head and neck Prostate	New: Gynecologic, thoracic, and abdominal organs Identical: Head and neck, prostate
Environment	The system can be used in a hospital environment or in a doctor's office.	The system can be used in a hospital environment or in a doctor's office.	Identical
JPEG image support	Yes	Yes	Identical
Import Treatment Plans	Yes. Import existing plans from third-party systems to compare dose objectives against templates.	Yes. Import existing plans from third-party systems to compare dose objectives against templates.	Identical

Element	Subject	Predicate	Conclusion
“Template” Treatment Plans	Yes. Factory-default plans with dose goals exist and users can configure a dose template.	Yes. Factory-default plans with dose goals exist and users can configure a dose template.	Identical
Automatic Initial Tumor Selection	Yes. Regions of interest are matched as the study is opened in the device. Users can adjust or match to more available regions of interest.	Yes. Regions of interest are matched as the study is opened in the device. Users can adjust or match to more available regions of interest.	Identical
Dose Objective Comparison	Yes. Comparisons can be done between more than one selected treatment plan. Dose is based on calculated dose and curated, gold-standard treatment plans.	Yes. Comparisons can be done between more than one selected treatment plan. Dose is based on calculated dose and curated, gold-standard treatment plans.	Identical
Image Viewer Capabilities	Yes. Display, pan, zoom, scroll, windowing, viewport layout.	Yes. Display, pan, zoom, scroll, windowing, viewport layout.	Identical
Calculate and Display Isodose Lines	Yes	Yes	Identical
Calculate and Display Dose Volume Histograms	Yes	Yes	Identical
Compare Dose from Multiple Plans	Yes	Yes	Identical
Dose Summation/ Treatment-Over-Time Data	Yes	Yes	Identical
Plan Review	Yes. Contains features for review of isodose lines, review of DVHs, dose comparison and dose summation.	Yes. Contains features for review of isodose lines, review of DVHs, dose comparison and dose summation.	Identical
Export Plan Information	Yes. Can export the selected plan for review and setup by a dosimetrist. Oncospace does not export a final plan, it will not export to a record-and-verify system.	Yes. Can export the selected plan for review and setup by a dosimetrist. Oncospace does not export a final plan, it will not export to a record-and-verify system.	Identical

VII. PERFORMANCE DATA

The following verification and validation testing results (performance data) support the substantial equivalence determination. Since this is a software-only medical device that does not control other devices the performance data does not include biocompatibility, electrical safety, electromagnetic compatibility, mechanical, acoustic, or animal testing.

The verification tests met all system requirements and acceptance criteria which address clinical, standard user interface, and cybersecurity requirements for the Oncospace device.

The validation testing for dose prediction was performed using retrospective clinical data. The Oncospace device's purpose is to reduce the effort needed to achieve a clinically viable and deliverable radiation treatment plan by supplying a treatment planning system with patient-specific plan optimization objectives derived from Oncospace dose predictions. Thus a trial of clinical performance was designed to demonstrate that plan quality, as represented by mean organ-at-risk (OAR) dose sparing, is non-inferior to that of plans created without use of Oncospace.. A comparison of target coverage was also made, because a comparison of OAR sparing is only valid if target coverage is maintained. Reduction in effort for the plans created with use of Oncospace was ensured by the planner using only the OAR optimization objective dose values supplied by Oncospace (rather than being allowed to adjust them via the usual trial-and-error process). Sample sizes were determined by estimating variance in mean OAR dose so that the trial would have 80% power at a significance level of 0.05 and a non-inferiority margin of 10 Gy. The head and neck and prostate models have previously undergone clinical performance testing, so for these only model performance testing (comparison of predicted dose values to ground truth values) was repeated here. Please note the distinction here between clinical performance testing (maintaining plan quality when Oncospace-derived objectives are used) and model performance testing (model accuracy).

Table 1 summarizes the characteristics of the datasets used for model development and internal testing, external performance testing, and clinical validation. Some datasets were fully decoupled from the medical record such that technique information could not be linked.

Anatomical Location	Development (Training/Tuning) and Internal Performance Testing Dataset (randomly split 80/20)	External Performance Test Dataset(s)	Clinical Validation Dataset
Head and Neck	1145 treatment plans for patients who received radiation therapy for HNC at Johns Hopkins University between 2008-2019. Plans were required to exhibit 90% target coverage. 10% of the plans were single-target and 90% were SIB (19% 2-target, 51% 3-target, 20% 4-target). 96% of plans had a total prescribed dose $\geq$ 60 Gy, 4% had a total prescribed dose < 60 Gy.	Dataset A: 265 patients who received radiation therapy for HNC at Institution_2. Plans were required to exhibit 90% target coverage. 11% of the plans were single-target and 89% were SIB (3% 2-target, 41% 3-target, 45% 4-target). 98% of plans had a total prescribed dose $\geq$ 60 Gy, 2% had a total prescribed dose < 60 Gy.	18 patients who received radiation therapy for HNC at JHU between 2021-2022. This included plans with 1-4 target dose levels, 10 to 48 fractions, in a variety of anatomical locations: lip, larynx, base of tongue, parotid, nasopharynx, scalp, neck, etc. HN was clinically validated in a previous 510k submission.

Anatomical Location	Development (Training/Tuning) and Internal Performance Testing Dataset (randomly split 80/20)	External Performance Test Dataset(s)	Clinical Validation Dataset
	The dataset contained 35% IMRT, 27% VMAT, 15% tomotherapy, 2% 3D conformal, and 22% unspecified plans.	Dataset B: 27 patients who received radiation therapy for HNC at Institution_3. Plans were required to exhibit 90% target coverage. All plans had 3 targets and total prescribed dose ≥ 60 Gy.	
Thoracic	1623 treatment plans (1437 lung and 186 esophagus) for patients who received radiation therapy for thoracic cancer at Johns Hopkins University between 2008-2019. Plans were required to exhibit 92% target coverage. 82% of plans had a total prescribed dose ≥ 45 Gy, 18% had a total prescribed dose < 45 Gy. 57% of plans had conventional fractionation (dose per fraction < 2.3 Gy) and 43% were hypo-fractionated plans.		20 patients (14 lung and 6 esophagus) who received radiation therapy at JHU between 2021-2024. This included targets in multiple locations within the lungs and esophagus, including single-target, SIB, and multi-phase courses, treated with conventional- and hypo-fractionation, and SBRT.
Abdominal	712 treatment plans for patients who received radiation therapy for pancreatic cancer at Johns Hopkins University between 2008-2019, and 69 treatment plans from patients who received radiation therapy for liver cancer at Montefiore Einstein Comprehensive Cancer Center. Plans were required to exhibit 85% target coverage. 3% of plans had conventional fractionation (dose per fraction		17 patients (11 pancreas and 6 liver) who received radiation therapy for at JHU between 2021-2024. This included single-target and SIB courses, treated with conventional- and hypo-fractionation and SBRT.

Anatomical Location	Development (Training/Tuning) and Internal Performance Testing Dataset (randomly split 80/20)	External Performance Test Dataset(s)	Clinical Validation Dataset
	<2.3Gy) and 97% were hypo-fractionated plans.		
Pelvis	1785 treatment plans (1662 prostate and 123 gynecological) for patients who received radiation therapy for pelvic cancer at Johns Hopkins University between 2008-2019. Plans were required to exhibit 94% target coverage. 98% of plans had conventional fractionation (dose per fraction <2.3Gy) and 2% were hypo-fractionated plans. 37% of plans had at least one phase with a nodal PTV, and 63% did not. The Prostate dataset used in the Pelvis model contained 15% IMRT, 46% VMAT, 14% tomotherapy, and 25% unspecified plans.	40 patients who received radiation therapy for prostate cancer at Institution_3. Plans were required to exhibit 94% target coverage. All plans had conventional fractionation.	17 patients (12 prostate and 5 gynecological) who received radiation therapy at JHU between 2021-2024. This included single-target and SIB courses, treated with conventional- and hypo-fractionation and SBRT. Prostate was clinically validated in previous 510k submission.

Table 1. Dataset characteristics.

Clinical performance testing involved comparison of plans generated with the aid of the Oncospace software to heterogenous sets of traditionally-planned clinical treatment plans.

Model performance testing involved comparison of DVH difference metrics with acceptance criteria.

- For the head and neck model, clinical performance testing was previously performed for the following OARs: brain, brainstem, spinal cord, left and right cranial nerve VIII (acoustic nerve), left and right parotid glands, left and right eyes, left and right lens, left and right optic nerve, optic chiasm, oral cavity, soft palate, glottis, cricopharyngeus, esophagus, sublingual gland, mandible bone, left and right submandibular glands, left and right cochlea, thyroid gland, and pharyngeal constrictor muscle(s).
- For the abdominal model, clinical performance testing was performed for the following OARs: bowel, duodenum, heart, left and right kidneys, liver, spinal canal, and stomach.
- For the thoracic model, clinical performance testing was performed for the following OARs: esophagus, heart, left and right kidneys, liver, left and right lungs, both lungs, spinal canal, stomach, and trachea.
- For the pelvis model, clinical performance testing for gynecological was performed for the following OARs: bladder, bowel, left and right femur heads, rectum, sigmoid colon. Previously,

clinical performance testing for prostate was performed for the following OARs: bladder, left and right femur heads, rectum, sigmoid colon, penile bulb, and bowel bag.

In clinical performance testing, for the plans using Oncospace, mean dose was statistically significantly lower for 5 OARs for abdominal and 4 OAR for pelvis (gynecological), and there were no statistically significant differences in mean dose for any of the remaining 11 OARs for thoracic, 3 OARs for abdominal, and 2 OARs for pelvis (gynecological). The trial demonstrated non-inferiority of mean OAR dose to 2.2 Gy for thoracic, 1 Gy for abdominal, and 1.9 Gy for pelvis (gynecological). There was no statistically significant difference in target coverage between clinical plans and plans created with use of the Oncospace system. Out of all the plans tested no plan required more optimization cycles using Oncospace versus using traditional radiation treatment planning clinical workflow.

All models met acceptance criteria for internal model performance. In external model performance testing for H&N plans, for Institution 2, mean absolute error in OAR DVH dose values is within 5% of the prescription dose value for 9/12 OARs, and does not exceed 9% for any OARs; for Institution 3, mean absolute error in OAR DVH dose values is within 5% of the prescription dose value for 10/12 OARs, and does not exceed 8% for any OARs. In external model performance testing for prostate plans, for Institution 3, mean absolute error in OAR DVH dose values is within 5% of the prescription dose value for 4/6 OARs, are 5.1% for one OAR, and 15.9% for one OAR. The small systematic differences between predicted and actual dose values are as expected given inter-institutional differences/preferences in contouring and in trade-offs between target coverage and OAR sparing.

The NLP model used for transforming non-standardized OAR names to standardized TG-263 names was trained using known name matches in the prostate, gynecological, head and neck, thoracic, and pancreas cancer datasets licensed to Oncospace by Johns Hopkins University. The model has undergone 5-fold cross validation, and external validation, for each anatomic region. During 5-fold cross-validation all models achieved a validation macro-averaged F1 score above 0.92 and accuracy above 96% for classifying previously unseen terms. The external testing dataset contained a total of 221 structures with 145 unique original names. The model correctly matched 207/221 (94.1%) of all structure names, or 131/145 (91.0%) unique structure names.

Conclusion

Verification and validation (including performance testing) was conducted in accordance with FDA guidance recommendations to confirm the device design met all specifications, user needs, and was acceptable to qualified clinical users. Oncospace has passed all the tests and the provided testing results demonstrate safety and effectiveness as compared to the predicate device. It is therefore concluded that Oncospace is substantially equivalent to the predicate device.

VIII. CONCLUSIONS

The subject Oncospace device is similar in intended use and functionality to the predicate Oncospace device. Oncospace has the same technological characteristics and features as the previously cleared device and does not raise new questions of safety or efficacy compared to the predicate device as demonstrated through the system design and testing.

Non-clinical and clinical verification, validation, and performance testing was conducted to confirm the device design met user needs and specifications and was acceptable to qualified clinical and non-clinical users. Oncospace has passed the verification and validation tests and provided clinical performance testing results with a library clinical dataset in order to demonstrate safety or effectiveness as compared to the predicate device. It is therefore concluded that the subject Oncospace device is substantially equivalent to the predicate Oncospace device.